

Microsoft Fabric Warehouse

Integration with Other Services

DBA Technical Reference | Volume 9 of 9

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1. Integration Architecture Overview

Fabric Warehouse sits at the center of the Fabric analytics platform, integrating natively with data ingestion tools, transformation engines, reporting layers, and external cloud services. All integration is mediated through OneLake and the Fabric SQL Analytics Endpoint.

Integration Category	Services
Reporting & BI	Power BI Desktop, Power BI Service, Excel, SSRS
Data Ingestion	Data Factory Pipelines, Dataflows Gen2, COPY INTO, Mirroring
Data Engineering	Spark Notebooks, Lakehouse, dbt
External Storage	Azure Data Lake Storage Gen2, Azure Blob Storage
Governance	Microsoft Purview, Fabric Admin Portal
External Databases	Azure SQL, SQL Server, Cosmos DB, Snowflake (via Mirror)
Developer Tools	SSMS, Azure Data Studio, Python (pyodbc), dbt, REST API

2. Power BI Integration

Fabric Warehouse auto-generates a Power BI semantic model from its schema, enabling instant report development without manual model building.

Connection Mode	Description	Best For
DirectLake	Reads directly from Delta Lake files — fastest BI performance	Large datasets, near-real-time reports
DirectQuery	Queries Warehouse SQL endpoint at report render time	Real-time data, row-level security
Import	Data imported into Power BI memory model	Small-medium datasets, offline access

DBA considerations for Power BI:

- DirectQuery reports generate T-SQL queries against the Warehouse — monitor these via DMVs
- Heavy Power BI concurrent usage consumes significant CUs — size capacity accordingly
- Apply Row-Level Security in the Warehouse (not just in the Power BI model) for server-side enforcement
- Use views in the Warehouse to pre-join tables — simplifies Power BI model and improves query performance

>> Create a Reporting View for Power BI

```
-- Pre-join fact and dimension for Power BI consumption
CREATE OR ALTER VIEW dbo.vw_SalesReport AS
SELECT
    f.OrderDate,
    c.CustomerName,
    c.Region,
    p.ProductName,
    p.Category,
    f.SalesAmount,
    f.Quantity
FROM    dbo.FactSales f
JOIN    dbo.DimCustomer c ON f.CustomerKey = c.CustomerKey
JOIN    dbo.DimProduct p ON f.ProductKey = p.ProductKey;
```

3. Data Factory Pipelines

Data Factory (Fabric Data Factory) pipelines orchestrate data movement and transformation into Fabric Warehouse. They support a wide range of source connectors and write modes.

Pipeline Activity	Use Case	Writes To
Copy Activity	Bulk data movement from 100+ sources	Warehouse table or Lakehouse
Script Activity	Execute T-SQL (stored procedures, COPY INTO)	Warehouse
Dataflow Gen2 Activity	Power Query transformations	Warehouse table
Notebook Activity	Spark-based transformations	Lakehouse or Warehouse via OneLake
Lookup Activity	Read data from Warehouse for pipeline branching	Reads from Warehouse
ForEach Activity	Loop over partition dates, tables, or files	Used to orchestrate other activities

>> Call Stored Procedure from Pipeline (Script Activity)

```
-- In the Pipeline Script Activity, set the query to:
EXEC dbo.usp_LoadDailySales @LoadDate = '@{formatDateTime(pipeline().parameters.loadDate, 'yyyy-MM-dd')}';
```

4. Dataflows Gen2

Dataflows Gen2 provide a Power Query-based transformation layer that can write transformed data directly into Fabric Warehouse tables. They are ideal for business-analyst-driven ETL without requiring T-SQL or Python.

Dataflow Feature	Details
Source connectors	300+ connectors including Excel, SharePoint, REST APIs, databases
Transformation	Power Query M language — same as Power BI Desktop
Destination	Fabric Warehouse table (direct write), Lakehouse
Write modes	Append, Replace, Merge (upsert)
Scheduling	Scheduled refresh or triggered by Data Factory pipeline
Staging	Optional staging Lakehouse for intermediate storage

5. Lakehouse Integration & Cross-Database Queries

Fabric Warehouse can query Lakehouse tables directly using cross-database T-SQL without any data movement. This is one of the most powerful Fabric integration features.

>> Cross-Database Query: Warehouse + Lakehouse

```
-- Query a Lakehouse table from within the Warehouse SQL editor
-- Syntax: [LakehouseName].[schema].[table]

SELECT
w.CustomerKey,
w.SalesAmount,
l.CustomerSegment, -- from Lakehouse
l.LifetimeValue -- from Lakehouse
FROM dbo.FactSales w -- Warehouse table
JOIN AnalyticsLakehouse.dbo.Customers l -- Lakehouse table
ON w.CustomerKey = l.CustomerKey
WHERE w.OrderDate >= '2024-01-01';
```

INFO: Cross-database queries within the same Fabric workspace require no additional authentication or linked server setup. The Fabric platform handles access automatically based on workspace permissions.

6. Azure Data Lake Storage (ADLS) Integration

>> Load from ADLS Gen2 Using COPY INTO

```
-- Load Parquet files from ADLS Gen2 into Warehouse
COPY INTO dbo.FactSales
FROM 'https://.dfs.core.windows.net//year=2024/month=06/*.parquet'
WITH (
FILE_TYPE = 'PARQUET',
CREDENTIAL = (IDENTITY = 'Managed Identity')
);
```

>> Load from Azure Blob Storage (CSV)

```
COPY INTO dbo.StagingOrders
FROM 'https://.blob.core.windows.net/raw/orders/2024/*.csv'
WITH (
  FILE_TYPE = 'CSV',
  FIRSTROW = 2,
  FIELDTERMINATOR = ',',
  CREDENTIAL = (IDENTITY = 'Managed Identity')
);
```

7. Azure Synapse Analytics Integration

Organizations migrating from Azure Synapse Analytics Dedicated SQL Pools to Fabric Warehouse can leverage the following integration and migration paths:

Migration Path	Description
Schema export	Use Synapse Studio to export CREATE TABLE scripts; run in Fabric Warehouse
Data export	Export Synapse tables to ADLS Gen2 as Parquet; COPY INTO Fabric Warehouse
CETAS migration	Replace Synapse CETAS patterns with Fabric COPY INTO + external tables
Workload migration	Move Synapse pipelines to Fabric Data Factory; repoint to Fabric Warehouse
Side-by-side running	Use Fabric cross-database queries to join Synapse and Fabric data during migration

8. OneLake Shortcuts

OneLake Shortcuts allow Fabric workspaces to reference external storage locations (ADLS Gen2, Amazon S3, Google Cloud Storage) without copying data. These shortcuts appear as Lakehouse tables and are queryable from the Warehouse.

Shortcut Target	Support Status	Access Pattern
Azure Data Lake Storage Gen2	GA	Direct file access via OneLake namespace
Azure Blob Storage	GA	Direct file access via OneLake namespace
Amazon S3	GA	Cross-cloud storage access
Google Cloud Storage	Preview	Cross-cloud storage access
Dataverse	GA	Microsoft Dataverse tables as Lakehouse shortcut

9. Microsoft Purview (Governance)

Microsoft Purview integrates with Fabric Warehouse to provide data governance, lineage tracking, data classification, and compliance management.

Purview Feature	Value to Fabric DBA
Data Catalog	Discover and document all Warehouse tables, views, and columns
Data Lineage	Visualize data flow from source systems through Warehouse to reports
Sensitivity Labels	Apply Microsoft Information Protection labels to sensitive Warehouse objects
Data Classification	Auto-classify PII, financial, and health data in Warehouse columns
Compliance reporting	Generate compliance reports for GDPR, HIPAA, SOC2 covering Warehouse data
Business Glossary	Link business terms to Warehouse objects for self-service discoverability

10. External Tools & Third-Party Connectors

Tool / Connector	Integration Method	Use Case
dbt (data build tool)	Fabric Warehouse dbt adapter (SQL endpoint)	SQL-based transformations and data modeling
Apache Airflow	REST API or SQL operator	Pipeline orchestration and scheduling
Fivetran	Native Fabric Warehouse destination connector	SaaS source data ingestion
Informatica	ODBC / JDBC connection	Enterprise ETL and data integration
Tableau	ODBC / native connector	Business intelligence and visualization
Qlik	ODBC connection	Business intelligence reporting
Python (pandas)	pyodbc / SQLAlchemy	Data science and custom analytics
R	RODBC / DBI packages	Statistical analysis
Excel	Power Query or ODBC	Self-service data access

>> Connect from Python (pyodbc)

```
import pyodbc
import pandas as pd

conn_str = (
'DRIVER={ODBC Driver 18 for SQL Server};'
'SERVER=.datawarehouse.fabric.microsoft.com;'
'DATABASE=;'
'Authentication=ActiveDirectoryInteractive;'
)
conn = pyodbc.connect(conn_str)

df = pd.read_sql(
'SELECT TOP 100 * FROM dbo.FactSales ORDER BY OrderDate DESC',
conn
)
print(df.head())
conn.close()
```